Ahmed Elsharkawy

Postdoctoral Researcher

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Objective

To contribute to impactful R&D initiatives by applying my expertise in human-centered design, AI-driven interaction systems, and human-robot interaction. I seek a dynamic environment where I can drive practical solutions, collaborate with interdisciplinary teams, and scale technologies for global impact.

Experience

DGIST InnoCORE Research Group for Bio-Embodied Physical AI (Affiliated with Daegu Gyeongbuk Institute of Science and Technology (DGIST) & Gwangju Institute of Science and Technology (GIST) | InnoCORE Postdoctoral Researcher **11 - 2025 ~ Present**

- Leading the research initiative entitled "Human-in-the-Loop Physical AI for XR-Based Digital Therapeutics"
- Leading multiple studies for SCI/SCI(E) journals, and top-tier conferences publications
- Supported teaching for graduate-level Human-Computer Interaction courses

Gwangju Institute of Science and Technology, College of Information and Computing, Department of AI Convergence, Human-Centered Intelligent Systems Lab. | Research Assistant Professor **10 - 2024 ~ 10 - 2025**

- Co-PI on highly competitive research grants
- Leading multiple studies for SCI/SCI(E) journals, and top-tier conferences publications
- Supported teaching for graduate-level Human-Computer Interaction courses
- Sharing research outcomes international academic conferences (e.g., CHI, UIST, Ubicomp, and ICRA)

Gwangju Institute of Science and Technology, School of Integrated Technology, Human-Centered Intelligent Systems Lab. | Postdoctoral Fellow **05 - 2023 ~ 09 - 2024**

- Leading and contributing to studies published in top-tier human computer interaction conferences (e.g., CHI and UIST)
- Integrated theory with real systems, conducting human-subject experiments
- Mentored junior researchers and interns, guiding team integration and idea development

Menoufia University, Faculty of Electronic Engineering, Department of Industrial Electronics & Control Engineering | Teaching Assistant **03 - 2014 ~ 08 - 2015**

- Taught practical labs for Microcontrollers, Embedded Systems, Power Electronics, Electrical Machines, Control Systems, and SCADA
- Mentored undergraduate students on graduation projects

Grand Solution for Technology (GST) / ADVANTECH | Automation Engineer (internship) **07 - 2012 ~ 12 - 2012**

- I trained in data acquisition systems and contributed to developing engineering solutions for civil and industrial applications.
- Gained experience in system integration and automation testing to ensure system efficiency.

Education

Gwangju Institute of Science and Technology, School of Integrated Technology (Currently College of Information and Computing, Department of AI Convergence) | Ph.D. in Integrated Technology (Graduate Program of Intelligent Robotics Technology) **09 - 2017 ~ 02 - 2023**

Research field: Robotics, Human-Computer Interaction, and Artificial Intelligence

Gwangju Institute of Science and Technology, School of Mechatronics (Currently School of Mechanical and Robotics Engineering) | MSc in Mechatronics Engineering 09 - 2015 ~ 08 - 2017
Research field: Robotics, Haptics, and Virtual Reality

Menoufia University, Faculty of Electronic Engineering, Department of Industrial Electronics & Control Engineering | BSc in Electronic Engineering 09 - 2007 ~ 07 - 2012
Major: Electronics **|** Minor: Control Engineering

Selected Publications

*** Equal contribution**

- [c.13] Gim, B., Kim, G., Yeo, D., Kang, Y., **Elsharkawy, A.**, & Kim, S. (**2026**). *From Disruption to Immersion: Reimagining Vehicle Motion as Environmental Feedback in In-Car VR*. In Proceedings of the Conference on Human Factors in Computing Systems (ACM CHI).
- [c.12] Ataya, A.*, **Elsharkawy, A.***, Lee, J., Hwang, S., Seong, M., & Kim, S. (2025). *ReD Shoes: Actuated Footwear for Multisensory Redirected Walking in Virtual Reality*. In Virtual Reality Springer.
- [c.11] Yeo, D., Kim, G., Oh, M., Park, J., Gim, B., Kang, S., **Elsharkawy, A,** & Kim, S. (2025). *AttraCar: Multisensory In-Car VR with Thermal, Airflow, and Motion Feedback through Built-In Vehicle Systems*. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (ACM UIST). [People's Choice Best Demo Award]
- [c.10] **Elsharkawy, A.**, Ataya, A., Yeo, D., An, E., Hwang, S., & Kim, S. (2024, May). *SYNC-VR: Synchronizing Your Senses to Conquer Motion Sickness for Enriching In-Vehicle Virtual Reality*. In Proceedings of the Conference on Human Factors in Computing Systems (ACM CHI). [Best Paper Honorable Mention]
- [c.9] Hwang, S., Oh, J., Kang, S., Seong, M., **Elsharkawy, A.**, & Kim, S. (2024, May). *Ergopulse: Electrifying your lower body with biomechanical simulation-based electrical muscle stimulation haptic system in virtual reality*. In Proceedings of the Conference on Human Factors in Computing Systems (ACM CHI). [Best Paper Honorable Mention]
- [c.8] Kang, S., Kim, G., Hwang, S., Park, J., **Elsharkawy, A**., & Kim, S. (2024, October). *Flip-Pelt: Motor-Driven Peltier Elements for Rapid Thermal Stimulation and Congruent Pressure Feedback in Virtual Reality*. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (ACM UIST).
- [c.7] **Elsharkawy, A.**, & Kim, M. S. (2022). *Human–Robot Labeling Framework to Construct Multitype Real-World Datasets*. IEEE Access, 10, 131166-131180.
- [c.6] Naheem, K., **Elsharkawy, A.***, Koo, D., Lee, Y., & Kim, M. (2022). *A UWB-based lighter-than-air indoor robot for user-centered interactive applications*. Sensors, 22(6), 2093.
- [c.5] Jeon, S., **Elsharkawy, A.**, & Kim, M. S. (2021). *Lipreading architecture based on multiple convolutional neural networks for sentence-level visual speech recognition*. Sensors, 22(1), 72.
- [c.4] **Elsharkawy, A.**, Naheem, K., Koo, D., & Kim, M. S. (2021). *A UWB-driven self-actuated projector platform for interactive augmented reality applications*. Applied Sciences, 11(6), 2871.
- [c.3] Ataya, A., Kim, W., **Elsharkawy, A.**, & Kim, S. (2021). *How to interact with a fully autonomous vehicle: naturalistic ways for drivers to intervene in the vehicle system while performing non-driving related tasks*. Sensors, 21(6), 2206.
- [c.2] Ataya, A., Kim, W., **Elsharkawy, A.**, & Kim, S. (2020). *Gaze-head input: Examining potential interaction with immediate experience sampling in an autonomous vehicle*. Applied Sciences, 10(24), 9011.
- [c.1] **Elsharkawy, A.**, Naheem, K., Lee, Y., Koo, D., & Kim, M. S. (2020, September). *LAIDR: A robotics research platform for entertainment applications*. In 2020 IEEE International Conference on Unmanned Aircraft Systems (ICUAS) (pp. 534-539).

Under Review Publications

*** Equal contribution**

- [c.14] **Elsharkawy, A.,** Gim, B., Ataya, A., Seo, Y., & Kim, S. (**2026**). *SelfBlending: Artificial Intelligence-driven Augmentation with Hand Interactions for Seamless Reality Blending in Virtual Environments*. In IEEE Transactions on Visualization and Computer Graphics.

- [c.15] **Elsharkawy, A.,** Ko, K., Kim, G., Choi, H., Seong, M., Ataya, A., & Kim, S. (**2026**). *SeatChat: Enabling Personalized and User-Configurable Sitting Feedback with LLMs*. In the Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT).

Selected R&D Experience

- [p.11] **Human-in-the-Loop Physical AI for XR-Based Digital Therapeutics**
  11 - 2025 ~ Present
  - Role: **Lead researcher**
  - Funding agency: InnoCORE program of the Ministry of Science and ICT (26-InnoCORE-01 and 26-InnoCORE-01)

- [p.10] **Development of dynamic real-virtual haptic feedback systems using personal robots**
  09 - 2024 ~ Present
  - Role: **Co-PI**
  - Funding agency: National Research Foundation of Korea (NRF)

- [p.9] **Bridging the gap between reality and virtuality: actuated XR systems powered by sensory intelligence and soft robotics** **05 - 2024 ~ Present**
  - Role: **Co-PI**
  - Funding agency: National Research Foundation of Korea (NRF)

- [p.8] **AI-powered intelligent surfaces for adaptive XR environments in vehicles**
  01 - 2024 ~ 10 - 2024
  - Role: **Project leader**
  - Funding agency: AI-based GIST Research Scientist Project, GIST Institute for Artificial Intelligence (GIAI).

- [p.7] **Human-AI Collaboration for XR Digitization of Physical Experience.**
  05 - 2023 ~ 12 - 2023
  - Role: **Project leader**
  - Funding agency: AI-based GIST Research Scientist Project, GIST Institute for Artificial Intelligence (GIAI).

- **Worked as a Research Assistant on several research projects:**
  - [p.6] GIST–MIT Research Collaboration (GIST, 05.2023–10.2024)
  - [p.5] AI-based Interactive Media for Cultural Heritage (NRF, 03.2020–03.2022)
  - [p.4] Multi-dimensional Filming for Immersive Performances (MCST & KOCCA, 03.2020–03.2021)
  - [p.3] Lighter-than-air Aerial Robot for Indoor Performances (KOCCA, MCST, 07.2019–12.2019)
  - [p.2] Robotics for Cognitive Improvement in Dementia (MOTIE, 02.2018–09.2018)
  - [p.1] Virtual Plant Training System (MOLIT & KAIA, 09.2015–08.2017)

Programming Languages and Engineering Software

- Python
- C++, embedded systems
- C#, Unity 3D
- Robot Operating System
- MATLAB & Simulink
- LabVIEW
- Ladder programming language, PLC (programmable logic controller)
- Electrical circuit design software (Eagle, Proteus, and Altium)

Honors and Awards

- **Selected for the Prestigious Global InnoCORE Postdoctoral Fellowship** (Supported by the Ministry of Science and ICT, Republic of Korea) **11 - 2025**
- **Best Paper Honorable Mention** at the conference on human factors in computing systems (CHI) 2024 **[c.10]** **04 - 2024**
- **Best Paper Honorable Mention** at the conference on human factors in computing systems (CHI) 2024 **[c.9]** **04 - 2024**

- **Excellence award for the outstanding performance at the AI-based GIST Research Scientist Project.** Awarded by the Artificial Intelligent school at Gwangju Institute of Science and Technology **12 - 2023**
- **Research achievement (RA) scholarship.** Awarded by the School of Integrated Technology at Gwangju Institute of Science and Technology **05 - 2018**
- **Republic of Korea (government/GIST) scholarship for doctoral studies 08 – 2017**
  Republic of Korea (government/GIST) scholarship for master's studies 08 – 2015

Professional Networking and Academic Profiles

- **LinkedIn**: Link
- **Google Scholar**: Link

Media and Press Interview

- **Walking Without Real-World Spatial Constraint, GIST Develops 'Red Shoes' to Overcome VR Walking Limitations** Press link: Press Release
- **Enjoying 4D VR in a Moving Car, GIST Develops Platform Enabling Immersive Experiences Using Only Built-in Vehicle Systems** Press link: Press Release
- **dmc channel, 'Logarithm' TV show**: **An overview of my research journey and discussion about robotics development** Episode link: YouTube
- **Two GIST doctoral researchers selected for the Ministry of Education's Academic Next Generation Support Project** Press link: Press Release
- **Human-Centered XR Experiences Break the Boundary between Reality and Virtuality** Press link: Press Release
- **GIST-MIT continues research results on 'human-centered' AI system created through close collaboration** Press link: Press Release